# ABIR KUMAR DAS

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## Senior Agentic AI Engineer | AI Architect | Enterprise AI & Backend Engineering

### Summary

- Senior Software Engineer and AI Engineering professional with 13+ years of experience designing, developing, and scaling enterprise-grade applications, backend platforms, microservices, and cloud-based solutions across healthcare and enterprise technology domains.
- Strong expertise in Java 17, Spring Boot, Python, REST API development, distributed systems, and enterprise application architecture, complemented by hands-on experience in Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and AI-powered automation.
- Experienced in building AI-assisted engineering workflows using LangChain, Spring AI, LLM integrations, structured outputs, and LLM response evaluation. Expanding capabilities in Agentic AI architecture, LangGraph-based orchestration, multimodal RAG, document intelligence, semantic retrieval, and reranking strategies to build context-aware and reliable AI applications.
- Knowledge of vector databases, embeddings, retrieval optimization, tool calling, API orchestration, and FastAPI-based AI service development.
- Combines deep enterprise backend engineering experience with modern AI architecture concepts to design scalable, secure, reliable, and production-ready AI solutions. Experienced in technical leadership, mentoring engineers, troubleshooting complex systems, and delivering enterprise-grade software.

### Technical Skills

**Programming Languages:** Java 17, Python, JavaScript, SQL

**Agentic AI & LLM Engineering:** Agentic AI, AI Agents, LangChain, LangGraph, LLM Integration, Tool Calling, Function Calling, Agent Orchestration, Structured Outputs, Prompt Engineering, Context Management, Multi-Step Reasoning

**RAG & Multimodal AI:** Retrieval-Augmented Generation (RAG), Multimodal RAG, Document Intelligence, Semantic Search, Hybrid Search, Embeddings, Vector Search, Retrieval Optimization, Reranking, Contextual Retrieval, Chunking Strategies

**Vector Databases:** OracleDB 23ai, Qdrant, ChromaDB, FAISS

**AI Frameworks & APIs:** FastAPI, Spring AI, REST APIs, LLM API Integration, Streaming Responses, API Orchestration

**LLM Evaluation & Guardrails:** Relevancy Evaluation, Fact-Checking Evaluation, Output Validation, Grounded Responses, Hallucination Reduction, AI Response Quality

**Backend & Architecture:** Java 17, Spring Boot, Microservices, Distributed Systems, REST API Design, JPA, Hibernate, Enterprise Application Architecture

**Databases & Data:** Oracle Database, Sybase, SQLite, Relational Data Modeling, Stored Procedures, Transaction Management

**Cloud & DevOps:** Oracle Cloud Infrastructure (OCI), Docker, Git, Maven, YAML, OpenAPI

**Testing, Security & Observability:** JUnit 5, Mockito, Fortify, OWASP Dependency-Check, OWASP ZAP, OpenTelemetry

## Work experience

**Oracle Cloud Infrastructure (OCI)** 2022-02 - 2026-10

**Senior Member of Technical Staff |Lead Backend & AI Engineering**

- Designed, developed, and enhanced enterprise-grade backend services using Java 17 and Spring Boot, supporting automated test-suite generation and intelligent engineering workflows.
- Built AI-assisted engineering workflows using Python, LangChain, and Spring Boot to automate data-context discovery, log interpretation, and integration-test workflow restructuring.
- Integrated Large Language Models (LLMs) into enterprise engineering workflows to improve automation, contextual understanding, and developer productivity.
- Implemented LLM response validation using Relevancy-Evaluator and Fact-Checking-Evaluator frameworks to assess generated content, improve factual accuracy, and enhance output reliability.
- Designed and maintained API-driven and YAML-based data-context workflows, including preview, save, read, update, and confirmation operations.
- Investigated and resolved complex concurrency and data-integrity issues involving optimistic locking, persistence updates, and Oracle Database transactions.
- Improved automated testing coverage using JUnit 5 and Mockito, strengthening application reliability and release quality.
- Collaborated across application, backend, and persistence layers to resolve complex production issues and deliver reliable enterprise solutions.
- Mentored 4+ junior developers and contributed to architecture discussions, code reviews, and engineering best practices.

**Cerner Healthcare** 2018-11 - 2022-02

**Senior Software Engineer | Backend Development**

- Developed and maintained Java and Spring Boot backend services for enterprise healthcare applications.
- Delivered application enhancements and defect fixes, participating in peer reviews and rootcause
  analysis to improve service reliability.
- Addressed security findings identified by Fortify, OWASP Dependency-Check and OWASP ZAP, supporting secure software development practices.

**GE Healthcare** 2014-05 - 2018-11

**Software Engineer | Application Development**

- Developed and maintained Java- and JavaScript-based healthcare applications supporting clinical and business workflows.

- Worked with Sybase databases and stored procedures to support transactional application functionality and data access.

**Foofys Solution and Tagtual Technology** 2013-01 - 2014-05

**Mobile Applications Developer**

- Developed mobile applications for Android and Windows Phone using Java, SQLite and PhoneGap.
- Integrated application features with HTTP and JSON-based services and web technologies.

## Education

**Kavi Kulguru Institute of Technology and Science, Nagpur University** 2008-09 - 2011-06

**Bachelor of Engineering**

Bachelor of Engineering (B.E.), Electronics & Communication

**Anjuman Polytechnic** 2005-08 - 2008-07

**Diploma**

Diploma, Electronics and Telecommunication -- Anjuman Polytechnic, Nagpur